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EVEN END SEMESTER EXAMINATION MAY- 2025

Name of the Course: **B.Tech**

Semester: **8**

Name of the Paper: **Digital System using VHDL**

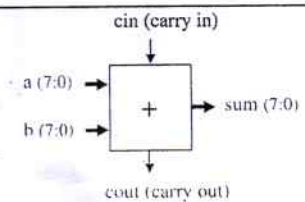
Paper Code: **TEC 853**

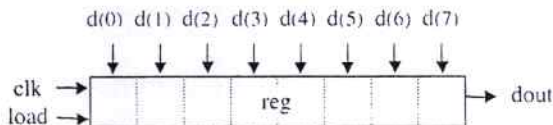
Time: **3 Hours**

Maximum Marks: **100**

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

Q1	(20marks)	
(a)	What is EDA? Discuss its significance in digital system design.	CO1
(b)	Discuss translation of VHDL code into a circuit.	
(c)	Explain ENTITY, ARCHITECTURE AND LIBRARY in VHDL with suitable examples of each?	
Q2	(20 marks)	
(a)	Explain pre-defined and user-defined data types in VHDL with suitable examples.	CO2
(b)	Discuss the various types of pre-defined operators in VHDL with examples.	
(c)	Explain the Signal Attributes and Data Attributes available in VHDL.	
Q3	(20 marks)	
(a)	 Using only concurrent statements, design VHDL code for unsigned adder of figure shown above.	CO3
(b)	Using process statement write the VHDL code for D-Flip Flop with Asynchronous Reset signal.	
(c)	Write the VHDL code to design 4-bit binary up/down counter.	

Q4	(20 marks)	
	Differentiate between the following: -	
(a)	i. Signals and Variables ii. WHEN and CASE statements	CO3, CO4
(b)	Explain PACKAGE in VHDL with example.	
(c)	What is a Finite State Machine? Design BCD counter using the concept of FSM.	
Q5	(20 marks)	
(a)	Discuss the 32-bit IEEE floating-point representation.	CO5, CO6
(b)	Design VHDL code for parallel to serial converter as shown below	
		
(c)	Write the VHDL code for the following FSM	
	